

MUSTASAARI VALASSAARET, FinlandWMO: **029100**Lat: **63.436N**Lon: **21.064E**Elev: **36**StdP: **14.68**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.4	4.8	-5.2	4.2	-0.2	0.8	5.7	5.0	36.7	23.3	33.8	23.3	9.5	140	0.764

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	72.3	66.2	69.4	64.4	67.0	62.5	67.4	71.5	65.1	68.7	63.1	66.4	7.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.6	94.6	70.3	63.4	87.6	67.8	61.5	81.7	65.5	31.8	71.8	30.1	68.7	28.6	66.6	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
33.0	29.0	25.6		-1.2	74.8	9.4	4.0	-8.0	77.7	-13.5	80.1	-18.7	82.4	-25.5	85.3
			DB	-1.9	68.1	8.8	2.8	-8.2	70.1	-13.4	71.7	-18.3	73.3	-24.7	75.2
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.0	25.2	22.6	27.7	35.3	43.9	53.0	60.7	60.5	52.7	42.9	35.5	31.0	
	DBStd	14.20	8.25	10.15	6.76	4.27	4.92	4.41	4.74	4.36	4.64	5.35	5.54	6.44	
	HDD50	4158	767	768	693	440	198	19	0	0	25	224	435	589	
	HDD65	8781	1232	1188	1158	890	654	362	153	152	368	685	885	1054	
	CDD50	884	0	0	0	0	9	107	331	325	107	5	0	0	
	CDD65	31	0	0	0	0	0	0	19	12	0	0	0	0	
	CDH74	47	0	0	0	0	0	0	30	17	0	0	0	0	
	CDH80	2	0	0	0	0	0	0	1	1	0	0	0	0	
Wind	WSAvg	12.7	14.1	13.3	12.9	12.0	11.7	11.1	10.0	10.9	12.4	14.5	14.8	14.9	
Precipitation	PrecAvg	20.3	1.5	1.1	1.2	0.9	1.3	1.6	2.4	2.2	2.0	2.2	2.2	1.8	
	PrecMax	26.9	3.0	2.3	2.0	2.4	2.9	3.8	5.0	3.9	4.1	4.2	4.4	3.4	
	PrecMin	15.2	0.2	0.0	0.3	0.1	0.2	0.5	0.2	0.6	0.6	0.5	0.4	0.4	
	PrecStd	3.5	0.7	0.7	0.5	0.6	0.7	0.8	1.2	1.1	0.9	0.9	0.9	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.8	38.6	40.6	49.9	61.0	68.5	77.6	76.1	65.0	54.9	47.8	42.3	
		MCWB	37.4	36.1	36.8	43.3	52.6	60.0	69.7	69.0	60.3	52.6	46.1	41.2	
	2%	DB	36.7	36.5	38.3	45.9	56.5	64.4	73.5	71.7	62.6	53.1	45.9	41.1	
		MCWB	35.3	34.8	35.5	41.3	50.1	57.8	67.2	66.1	58.8	51.4	44.6	39.8	
	5%	DB	35.8	34.9	36.8	43.4	53.5	61.9	70.6	68.9	60.7	51.4	44.3	39.8	
		MCWB	34.5	33.7	34.7	39.8	48.0	56.5	65.4	64.1	57.5	49.9	43.2	38.6	
	10%	DB	34.5	33.7	35.5	40.9	50.9	59.8	68.1	66.9	58.9	49.8	42.7	38.5	
		MCWB	33.5	32.9	33.8	38.0	46.5	55.2	63.6	62.5	56.2	48.0	41.6	37.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.9	36.7	37.8	44.7	54.2	61.8	71.0	70.5	61.6	53.6	46.6	42.0
			MCDB	38.2	38.3	40.0	48.6	58.9	66.4	75.6	74.8	64.1	54.5	47.4	42.3
2%		WB	35.7	35.2	36.0	42.0	50.8	58.9	68.4	66.6	59.6	51.8	45.0	40.3	
		MCDB	36.3	36.3	37.6	45.3	55.4	63.0	72.7	70.5	61.8	52.8	45.8	40.9	
5%		WB	34.7	34.1	35.0	40.1	48.6	57.3	66.2	64.8	58.2	50.1	43.4	38.8	
		MCDB	35.5	34.7	36.4	43.1	52.7	61.0	69.9	68.3	60.2	51.3	44.1	39.5	
10%		WB	33.6	33.0	34.0	38.3	47.0	55.9	64.0	63.2	56.7	48.4	41.9	37.4	
		MCDB	34.5	33.7	35.3	40.7	50.4	59.2	67.6	66.4	58.5	49.6	42.6	38.3	
Mean Daily Temperature Range	5% DB	MDBR	6.7	7.1	7.5	7.3	8.6	8.6	8.2	7.7	6.5	4.9	4.8	5.3	
		MCDBR	5.8	5.8	7.0	10.9	12.9	11.8	11.1	10.2	8.4	5.4	4.2	5.1	
		MCWBR	5.6	5.4	5.5	7.0	7.8	7.1	6.4	6.3	5.9	4.8	4.3	5.0	
	5% WB	MCDBR	5.3	5.6	6.5	10.4	11.8	10.5	10.3	9.4	7.7	5.0	4.2	5.0	
		MCWBR	5.3	5.3	5.3	7.1	7.9	6.9	6.4	6.2	5.9	4.7	4.5	5.1	
Clear-Sky Solar Irradiance	taub	0.233	0.289	0.312	0.351	0.352	0.351	0.371	0.360	0.333	0.316	0.372	0.350		
	taud	2.040	2.154	2.214	2.279	2.448	2.498	2.466	2.492	2.542	2.384	2.645	2.570		
	Ebn at Noon	118	199	246	262	274	276	266	259	245	193	109	62		
	Edh at Noon	10	20	28	33	31	30	30	27	21	16	9	5		
All-Sky Solar Radiation	RadAvg	43	224	706	1220	1659	1879	1714	1300	758	285	68	21		
	RadStd	5	45	83	118	153	146	153	114	72	55	13	2		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page